



Scrödinger's Data in OpenStreetMap

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Who has mapped one of these?



- As maxspeed on a way?
- As traffic-sign=maxspeed on a node?
- Both ways?
 - If yes is the ratio between the two closer to 1:1 or 1:100?

Ubiquity

- Data not information
- Unknown knowns
- Embedded data
- Emergent



Nightingale



TOPICS IN DATAVIZ

The Bus Station That Didn't Exist, and Other Data Epiphanies

Giuseppe Sollazzo • July 29, 2025



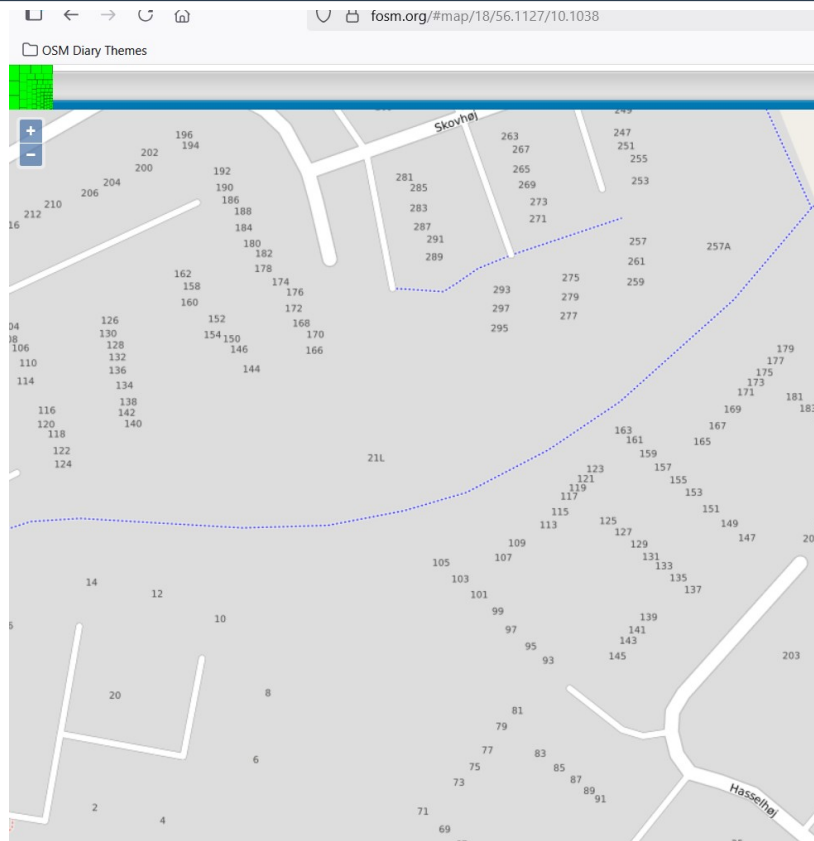
"Data is multidisciplinary" is my mantra
—it's 2025, and I've now worked 20 years

Why?

- **Data not being used for its original purpose**
- **Support for common, but not rare, use-cases**
- **‘Fragmentation’ : lose some value from previously aggregated data**
- **Failure to collect data**

OSM Perspective

- **POTENTIALLY a universal geodatabase**
 - Not so in practice
- **Human scale from 1 metre to ~ a few kilometres**
- **Favoured Use-cases**
- **Data getting more granular**
- **Imports and specific mapping projects**
- **Evolution of mapping/tagging practice**
- **Variation of mapping by location**
- **Patchiness (anisotropy) at all scales**
- **Fuzziness of some mapped elements**
- **Mapper demographics**



Scope of remainder of talk

- **What kinds of additional data can be derived using OSM alone?**
 - **Out of scope**
 - Solution through additional tags/relations etc
 - Enrichment of OSM from external sources
 - **Much Reduced scope**
 - Tag-only examples
 - Routing
 - Geocoding
 - Cartography

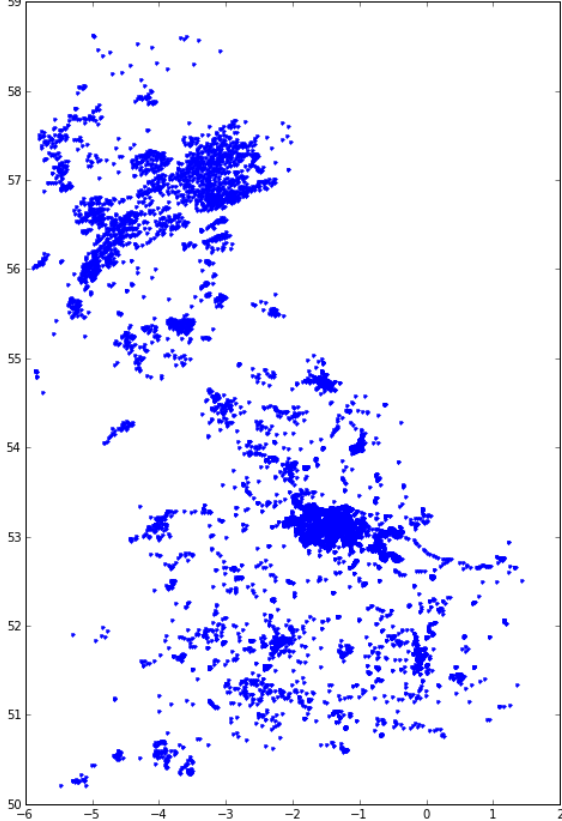
Tag only

- Mapping tags to more formal, use-case specific, schemas
- **Basic Enrichment**
 - Implicit tags become explicit
 - Add geolocation codes
 - Add hierarchies
- **Harmonising variant tagging schemes**
- **Tools**
 - LUA
 - Osmosis tag transform
 - Osmium
 - Limited capability for small examples
 - Overpass
 - Limited capability for small examples
 - GIS Applications, Packages & Libraries (QGIS, Postgres/PostGIS, R, geopandas ...)

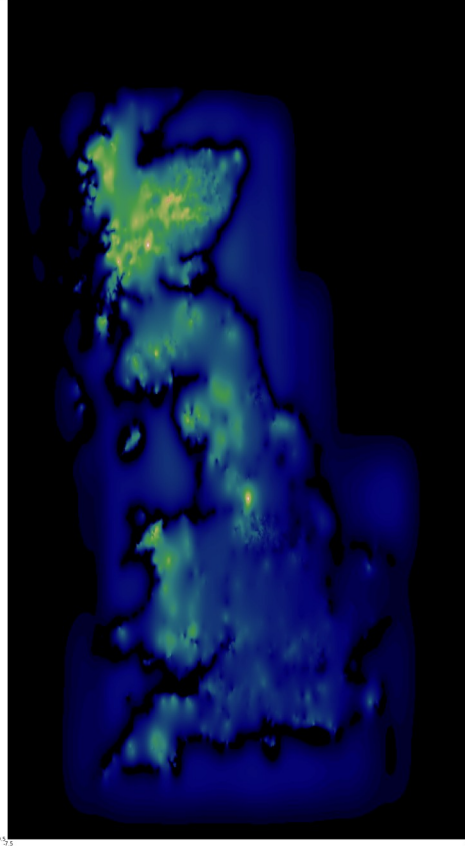
```
-- -----  
-- Shopmobility  
-- Note that "shop=mobility" is something that _sell  
-- handled as shop=nonspecific for now.  
-- We handle some specific cases of shop=mobility he  
-- -----  
if (( passedt.amenity == "mobility"  
    ( passedt.amenity == "mobility_equipment_hire"  
    ( passedt.amenity == "mobility_aids_hire"  
    ( passedt.amenity == "shop_mobility"  
    (( passedt.amenity == "social_facility"  
        ( passedt.social_facility == "shopmobility"  
    ((( passedt.shop == "yes"  
        ( passedt.shop == "mobility"  
        ( passedt.shop == "mobility_hire"  
        ( passedt.building == "yes"  
        ( passedt.building == "unit"  
        (( passedt.name == "Shopmobility"  
        ( passedt.name == "Shop Mobility"  
    passedt.landuse = "unnamcommercial"  
    passedt.amenity = "shopmobility"  
end
```

Gaussian / Kriging (smoothing patchy/noisy data)

OpenStreetMap elevation point data for Britain. (This image is (c) Dan Stowell, CC-BY. The underlying data is (c) OpenStreetMap contributors, ODbL.)

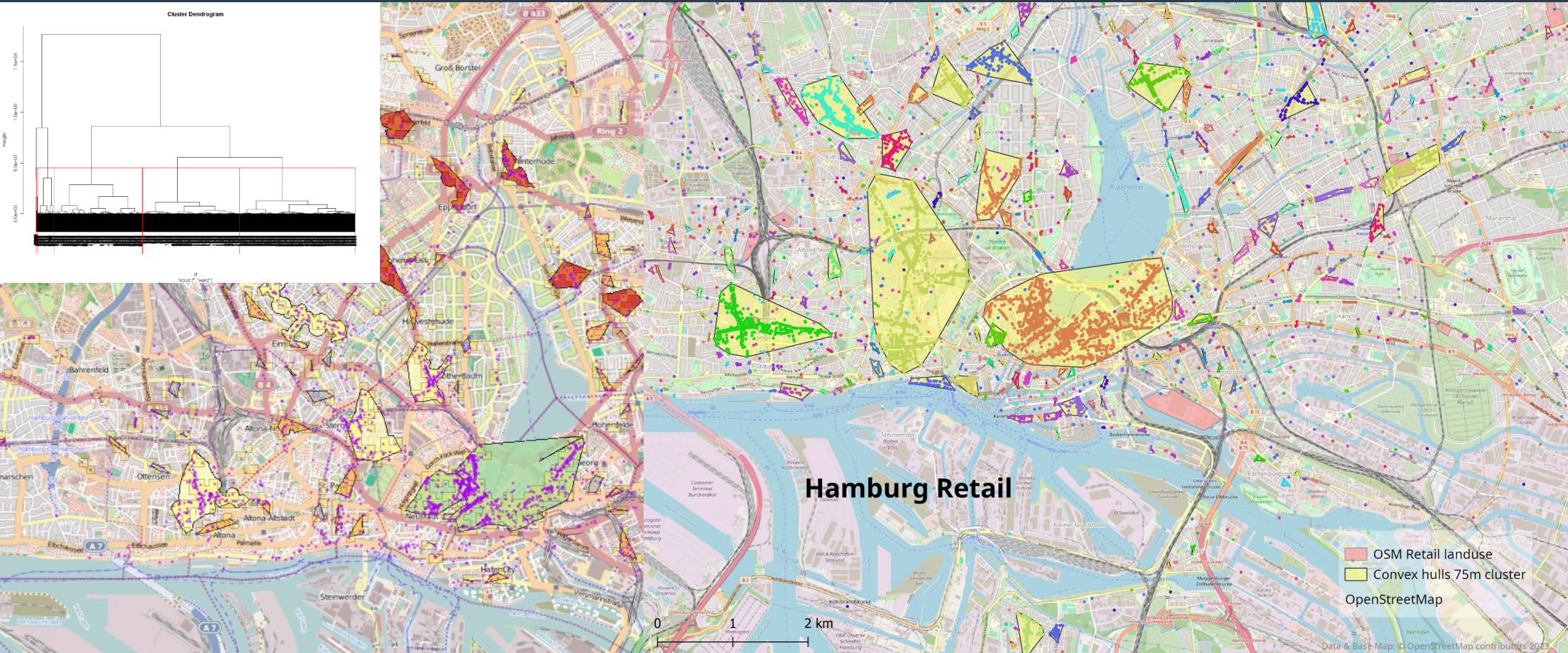


Reinterpolated map of Britain, estimated from OpenStreetMap point data, using Gaussian Processes (GP). (This image is (c) Dan Stowell, CC-BY. The underlying data is (c) OpenStreetMap contributors, ODbL.)

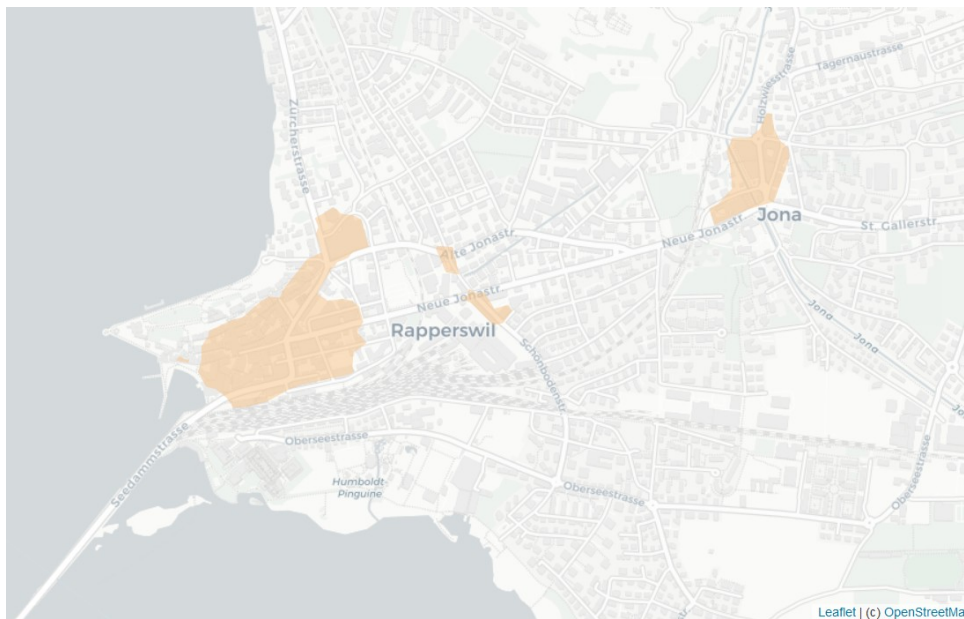


- **Work of Dan Stowell**
- **Used ele tags across Great Britain**
 - Not just on peaks
 - Often in data derived from GPS tracks
- **Demonstrates principle**
 - Not particularly useful

Retail Areas from Shop Locations (Germany)



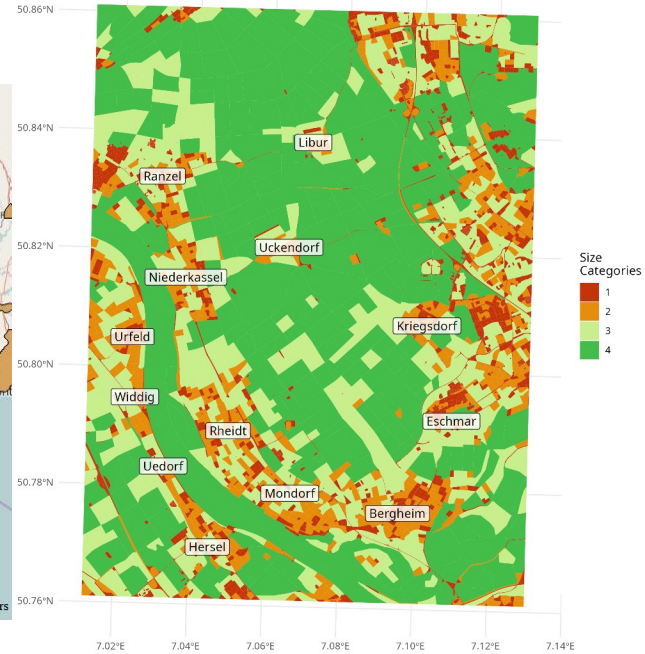
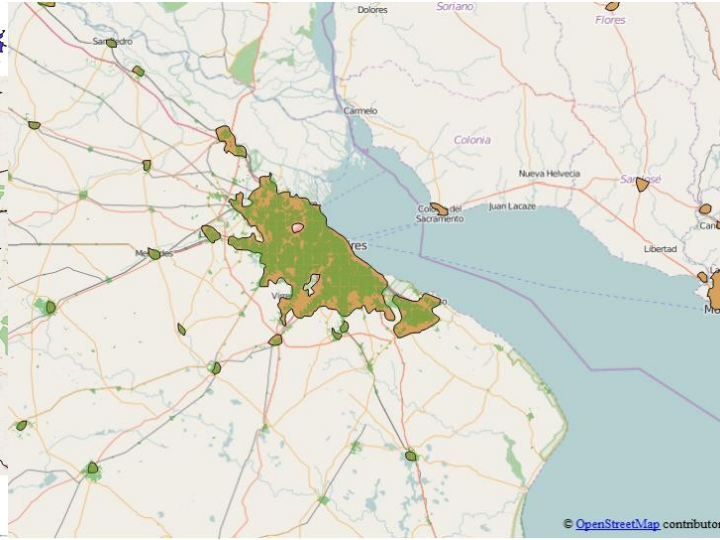
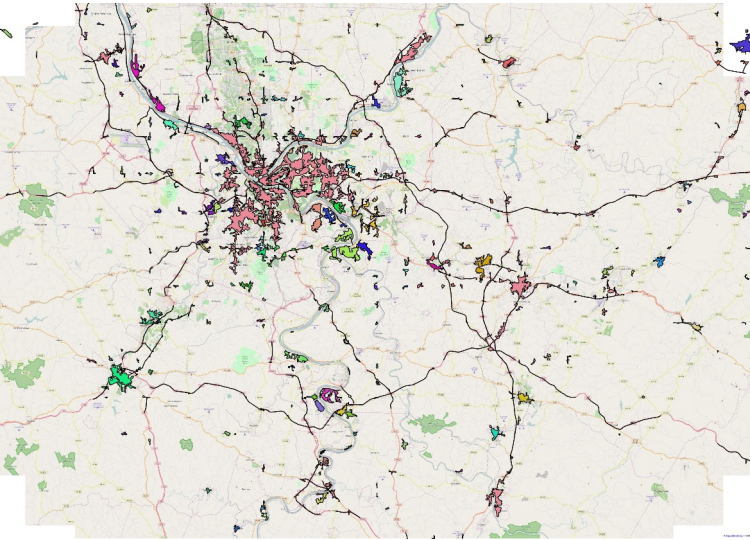
Areas of Interest



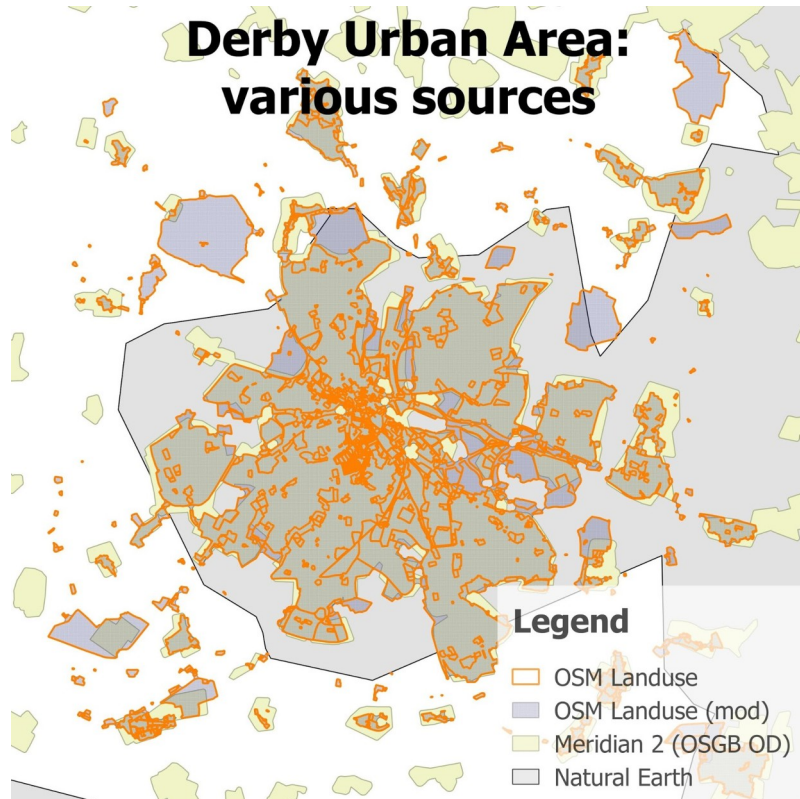
Work of Prof. Stefan Keller and colleagues at Rapperswil.
Presented at SotM-18 Milano:
https://2018.stateofthemap.org/2018/A30-Areas-of-Interest_for_OpenStreetMap_with_Big_Spatial_Data_Analytics/

- **Buildings containing 80+ different retail, hospitality & tourism POIs**
- **Clustered with DB Scan**
 - Parameters are localise
- **Create Concave Hulls**
- **Apply network centrality of closeness to street grid**
- **Sanitize polygons**

Urban Areas 1

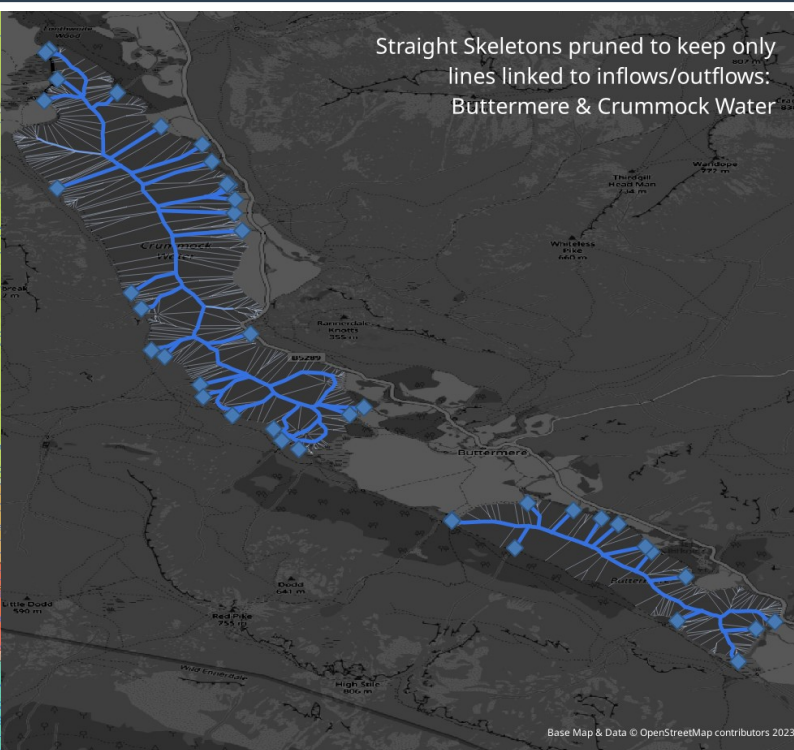
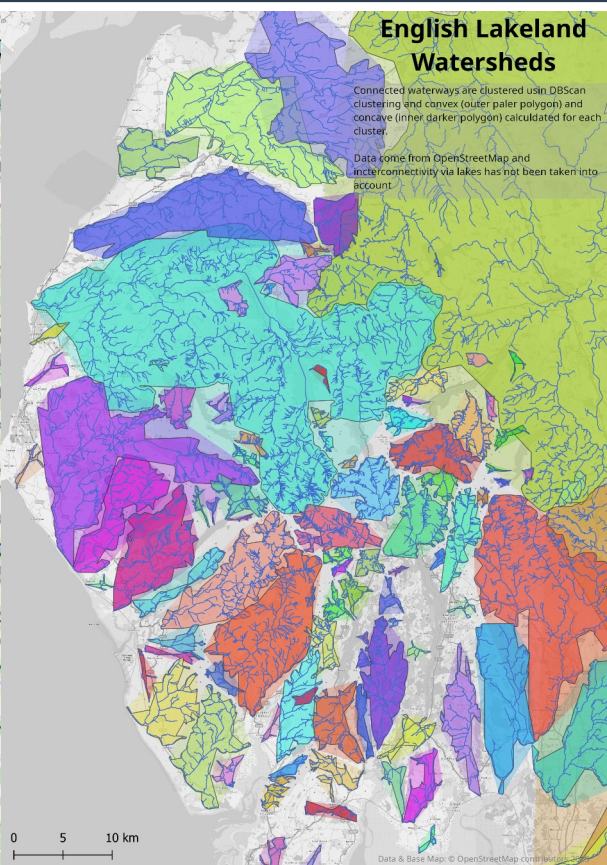
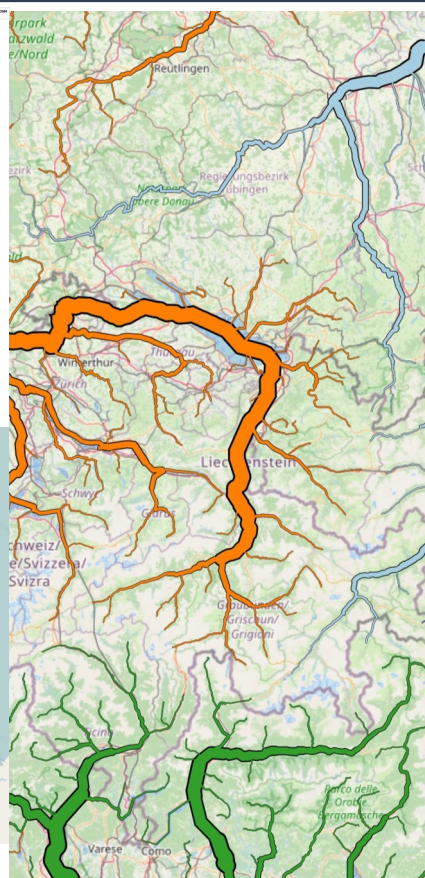
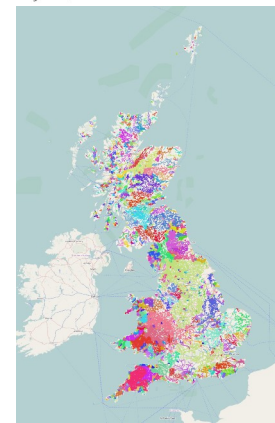
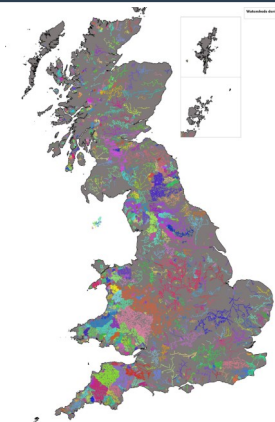


Urban Areas 2

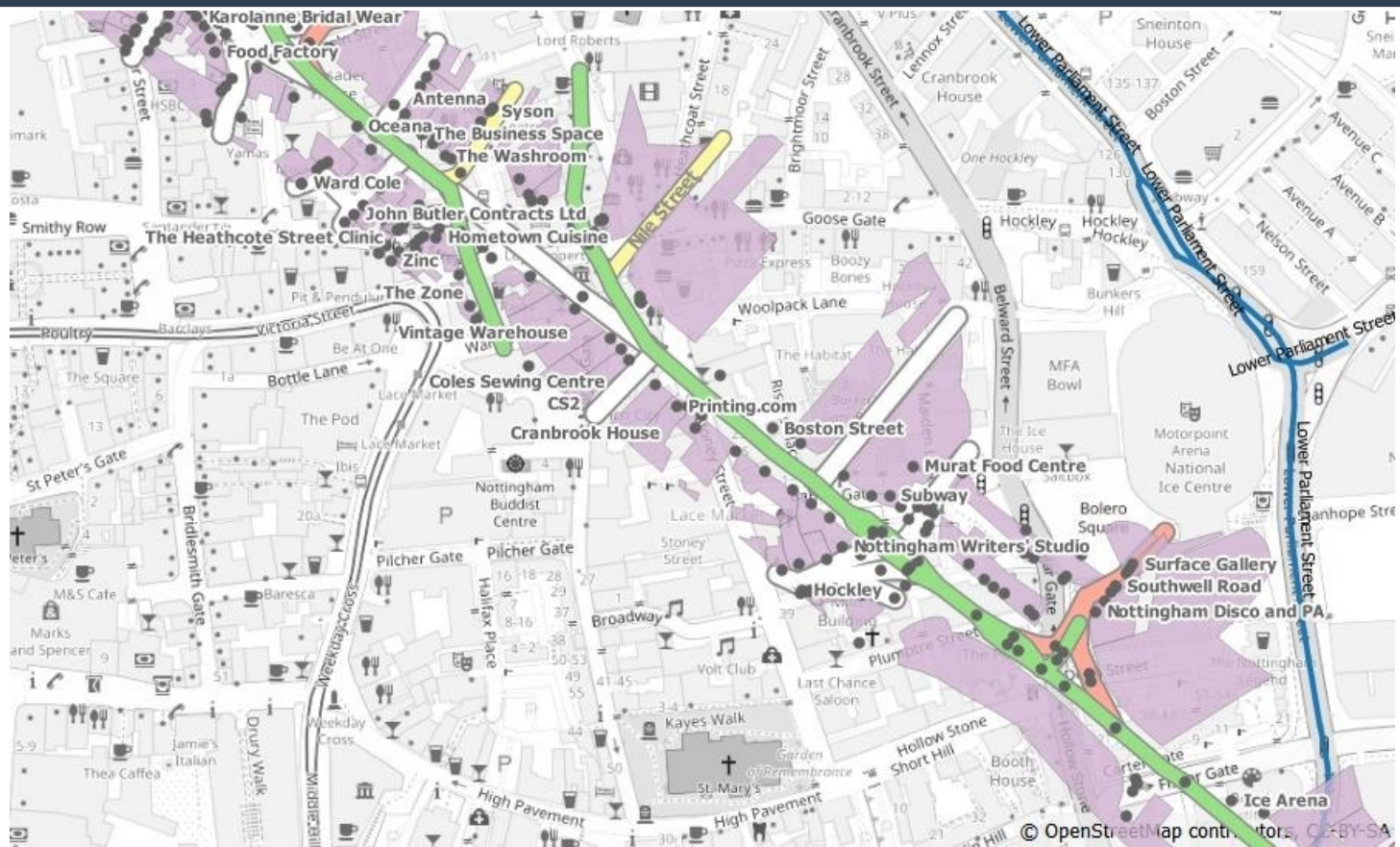


- **Using the road network**
 - Short road segments
 - Road density in a grid
 - Polygon method
- **Landuse aggregation**
- **POI clustering**
 - Not tried
- **Conflation of multiple approaches**
 - Not tried

Waterways & Hydrographic Basins



1-D Maps



1 St Paul's Cathedral (*place of*)

84  The Auld Tram (*cafe*)  
 Optical Express (*optician*) 

.. – [disused shop]

80 SD Greggs (fast food) F B W
 -- Ladbrokes (bookmaker) B

-- Slaters (clothes) ☐ B ☐ W

78 **SD** – [*disused shop*]

77 MERKUR Slots (*adult gaming*)

75  Black Sheep Coffee (cafe) [

71 -- Dundee Backpackers Hostels

2-73 SD Robertson & Watt (*jewelry*) [

● 三、(1) 100 元 (2) 100 元 (3) 100 元

-- Timpson (*locksmith*) B W

R-69 Bayne's (bakery)

09 ● Baynes (baker)) []

-- CK Electronics (mobile phor

5-66 SA PricePoint (variety store) F

-- Maisie & Mac [*disused gift*]

-- RBS (*bank*) ☐ B ☐ W

5-6 **SS** Morrisons Daily (convenienc

Virgin Money (bank) B

 Dundee Farmers' Market (m)

Tools & Techniques

"Buffers are your friend" Melaskia to me on IRC 2009

- **Way and Graph Chaining**
 - e.g. Amanda's lump ways
- **Skeletonisation & Midlines**
- **DB Scan Clustering**
- **General Polygon manipulation**
 - Buffers
 - Hulls
 - Union/Intersect
- **'Fake' OSM data**
- **Divide & Conquer approaches**
 - Gridding data
 - Convert to a non-geospatial problem
 - Restore geometries at the end
 - Particularly useful for graphs
- **Locally Contextual Parameterisation**
- **Defensive approaches**
 - Plan for loss of data
 - Invalid Polygons; sharding
 - Alternative techniques
- **Making the Data Available**

Challenges

- **Address Interpolation**
- **Non-overlapping buffered Coastlines**
- **Sidewalks from sidewalk=***

Resources & Learning More

- **Blog Posts**
 - Maps Matter
 - <http://sk53-osm.blogspot.com/>
- **#30DayMapChallenge**
 - Often creative use of OSM data
- **Previous State of the Map Talks**
 - Links on relevant slides
- **Project Websites**
 - Linear Maps : Robert Whittaker
 - Waterway Maps : Amanda McCann